The Impact of Biases in Facial Recognition Artificial Neural Networks

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Abstract

This study probes how biases are formed and mitigated within artificial neural networks used for facial recognition. In current research on facial recognition neural networks, there are many ways that bias can negatively affect the accuracy of the network on characteristics such as gender status and identity. In order to test this, two pre-trained neural network models were fed two novel datasets - one on cisgender faces and one on transgender faces. The two pre-trained models were then analyzed using logistic regressions for gender status and identity variables on accuracy rates from prediction outputs. There were main effects of model bias based on model, gender identity, and gender status, and interactions based on model and gender status.

Introduction

Neural Network Architecture

Artificial neural networks (ANNs) are a subset of artificial intelligence and machine learning that can provide us with predictions on the information we care about. They are based on human neural networks located within our brain, that consists of neurons, synapses, and electrical impulses (Gupta, 2013). To put it simply, neural networks consist of inputs and outputs. The inputs, which can be any type of information, are fed into the neural network (NN) in order to make it "learn" and can be used to "test" how well the algorithm performs after it has "learned" from data. This "learning" is achieved through shifting the weights in the artificial neurons within the neural network, which simulates the human process of learning and facilitates the "conversations"[1] that happens between neurons. These "conversations" form connections between neurons, sometimes across multiple layers (called a hidden layer[2]), that will then converse with each other to form predictions and share outputs which serve as the output. The resulting outputs are a form of transformations from the original inputs (DeepAI, 2019). For the scope of this paper, the inputs will be images of cisgender and transgender people with different demographics (e.g., age, race), and the output will be a prediction of their gender identity (e.g., man, woman). The working definition of cisgender at the time of publication is someone who identifies with their assigned gender at birth, and the definition of transgender is someone who does not identify with their assigned gender at birth.

Neural networks must be trained before testing and accessing outputs. Datasets used for training should be relevant to desired outputs (i.e., for outputs on human faces, datasets should include human faces). Training times will typically scale according to database size, with larger datasets causing longer training requirements. Training on too large of a dataset can lead to an unbalanced neural network, especially in Deep Learning NN architectures with larger hidden layers. This is known as 'overfitting,' which may detrimentally impact future predictions and outputs when testing NNs (Guo et al. 2015).

After training a neural network, the weights will have shifted to accommodate the dataset it 'learned' from. With each new input, the neural network will change its neurons' weights, similar

[1] "Conversations" denote that one neuron or node within the algorithm will share information to similar nodes, forming connections.

[2] A hidden layer consists of one or more layers in between the input and output layers. Hidden layers allow for increased connections during inter-node "conversations," increasing output accuracy.

to learning within the human brain. The weights set during the training phase will be used to create the predictions used as outputs. Weights are extremely important to how a network can function – both in humans (via action potentials[3]) and in machines. In both cases, the weights determine if a singular neuron activates or not when thought based on a set threshold. Neurons with higher thresholds have stronger weights and increased potential to fire, with the inverse being true for lower threshold neurons. The actual threshold representing the weight is arbitrary, whereas the magnitude of the threshold determines neuron firing. Neurons will only activate if the cumulative weight is greater than the set threshold; activation will not occur if there are not sufficient excitatory or inhibitory signals (DeepAI, 2019).

The most basic example of a neural network is a single-layer perceptron. Although perceptrons were originally introduced by Rosenblatt (1958), they were popularized by two computer scientists, Minsky and Pappert, in 1969 after publishing their book *Perceptrons*. Single- and multi-layer perceptrons are the predecessors to modern Artificial Intelligence (AI) algorithms like Artificial Neural Networks (Bhardwaj, 2020). Neural Network learning can be readily explained by an "*AND*" perceptron. The *AND* perceptron takes 2 inputs, and has 1 output node. Both of the input values (input one *AND* input two) must be more than 0 to fire. If both input weights are 1 and the threshold 1.89 the neuron is likely to fire, due to the magnitude of the weights being higher than the threshold. Any inputs that do not exceed the threshold (e.g., an input of 0 and 1 or 1 and -1) will not cause activation.

For the scope of this project, two pre-trained Convolutional Neural Networks (CNNs) were used. Convolutional Neural Networks rely on convolution to compute output predictions, whereas Artificial Neural Networks rely on the traditional weight and node system described above. CNNs use both feature extraction and feature mapping during computation, while ANNs typically only extract features. For feature mapping, CNNs will use a specialized "convolutional layer" within the hidden layer to extract relevant information from the inputs, which are then transformed into a map onto the output image. Convolutional layers will be followed by pooling layers iteratively, reducing the dimensions and preserving important information until the end of the hidden layer. A final, fully-connected layer is typically used in order to convert the two dimensional features that have been fed forward into a one dimensional

[3] Action potentials are the biological equivalent to the ANNs' weights. After the amount of excitatory or inhibitory energy exceeds the threshold, the neuron will fire. If there is not sufficient energy, the neuron will not fire.

vector that will supply the output that is readable to humans. The fully-connected layer contains around 90% of the parameters from the entire CNN model architecture (Guo et al., 2015), influencing the final model outputs.

Convolutional Neural Networks are often used for computer vision[4] due to their complexity. An example of an output from a computer vision NN is detecting a face within an image, such as what was done within the algorithm used for this study. Additional things that can be accomplished with computer vision include, but are not limited to: image classification[5] and facial recognition, detection, and classification[6]. This study was focused on facial recognition, specifically model-predicted classification of gender identity based on facial feature extraction.

Algorithmic Bias

Transgender and non-White individuals have historically been underrepresented in research, especially in the field of Artificial Intelligence (AI). Since the first example of artificial intelligence in 1958, technological advancements have exponentially increased algorithmic capacity. Despite these advancements, conversations about ethics have lagged far behind. According to the AI Index 2022 Annual Report (Zhang et al., 2022), the topic of ethics within machine learning started to be discussed more prevalently circa 2014. Additionally, the reported number of publications on the subject of AI ethics have increased by a factor of 5 since 2014, with 71% of the publications pertaining to organizations and corporations in industry. Following this spike, several projects on machine learning (ML) ethics launched in 2017, including the *Pan-Canadian Artificial Intelligence Strategy* and *AI Technology Strategy* (EPRS, 2020) and MIT's *Ethics and Governance of Artificial Intelligence* (MIT Media Lab, 2017). This spike in ethical discussions of AI led to publications on facial recognition, such as Dr. Buolamwini's dissertation on harmful intersectional biases in neural networks (Buolamwini & Gebru, 2018).

Transgender individuals and Black people - specifically Black women - are most affected by unchecked algorithmic bias. Buolamwini and Gebru's *Gender Shades* (2018) found that darker-skinned Black women are more likely to be misgendered by computer vision software. In *How Computers See Gender* (2019), Scheuerman et al. found that transgender people are also often misgendered by computer vision algorithms. Despite the documented dangers and problems

[4] There is a subset of NN algorithms which are trained on 'Computer Vision' tasks, like taking image, video, or web camera feedback. Computer Vision NNs have higher accuracy when processing 'visual' inputs compared to other NN subtypes.

[5] An example of image classification is AlexNet; with an input image of a bird, the output is likely to be 'bird.'

[6] Detecting a face within an image, recognizing a specific individual's face, and classifying certain characteristics of the face like gender, race, age, etc. are all things that Computer Vision algorithms can do with respect to facial analysis.

associated with facial recognition software, machine learning and neural networks are rapidly gaining popularity. Concerns regarding use of Automatic Gender Recognition (AGR) NNs can adversely affect transgender populations and those who did not consent to data usage, especially if users are not allowed to change their algorithmically-assigned gender (Keyes, 2018; Scheuerman et al., 2019, 2020). Biases in facial recognition neural network algorithms are not exclusive to transgender and Black woman populations; there have been instances of law enforcement wrongfully arresting Black men after misidentification (Johnson, 2022). Some government programs have also been found to use AI in ethically questionable ways, such as the police in Detroit, Michigan (Johnson, 2022), England, and Wales (Radiya-Dixit, 2022).

In addition to questionable ethics, transgender and non-White populations are typically underrepresented in the datasets used to train and test the algorithms (Wu et al., 2020 ; Karkkainen & Joo, 2021). There have been attempts in the past to create datasets to help NNs train on specific populations. To train facial recognition algorithms on transgender people, the HRT Trans Database (AIAAIC, 2023) was created but heavily criticized for questionable ethics: lack of information on consent from people in the photos, lack of publication of the dataset despite being publicly funded, and disputes over the dataset being created for “national security purposes” (AIAAIC, 2023). Targeting of marginalized communities have led to increased ethical and privacy concerns when publicly releasing datasets. To avoid underfitting a model, neural networks should be trained on diverse demographics through balanced datasets; machine learning models that have not been trained on non-White and transgender populations risk reduced accuracy on said groups. Balanced datasets contain near proportionate amounts of images consisting of faces from each demographic to be measured during testing and deployment. Many readily available large-scale training sets contain disproportionate (unbalanced) levels of cisgender white men, directly contributing to model underfit for diverse populations. These commonly used datasets include LFW (Labeled Faces in the Wild), CASIA-WebFace, MS-Celeb-1M, Adience, and VGG Face datasets (Papers with Code, 2023). To mitigate this issue, Karkkainen & Joo (2021) created a novel database for training and testing facial recognition algorithms. The “FairFace” dataset boasts images with diverse racial identities, genders, and ages, with racial balancing. Depending on the use case, unbalanced datasets may be more useful for facial recognition algorithms. Compared to larger-scale datasets like LFW, the FairFace dataset cannot be easily split between training, validation, and testing datasets. Each algorithm should use datasets that fit within project resources, while also maintaining optimal model fitness. Although pre-created datasets may be convenient, unbalanced demographics can lead to unfit models and detrimentally impact equity and accuracy when used on diverse populations.

Although unbalanced datasets can contribute to poor model fit, not all biases that come from AI algorithms are formed during the training process. Researchers have sought to determine how biases form, and mitigation methods before detrimental harm comes to end-users (Wu et al., 2020; Scheuerman et al., 2020; Buolamwini & Gebru, 2018; Google, 2022). Balancing datasets is one proposed way to debias algorithms (Joo & Kärkkäinen, 2021; Wu et al., 2020), while others have argued that balancing datasets will not be enough for bias mitigation (Wang et al., 2019A ; Wang et al., 2019B; Alberio et al., 2020; Gong et al., 2020; Zhang et al., 2018).

The current study sought to probe debiasing methods relating to gender identity and gender status within facial recognition neural networks. Gender identity and gender status were defined using binary man-woman and cisgender-transgender dichotomies, respectively. The performance of two novel datasets from scraped images of cisgender and transgender faces were compared between pre-trained models on either balanced (FairFace) or unbalanced (InceptionResNet v1) training datasets. It was questioned if FairFace's dataset balancing would impact accuracy rates within transgender populations. Additionally, this study sought to understand how the balancing of training datasets would impact performance on gender classification outputs.

Methods

Models

In order to find and mitigate biases, two pre-trained neural network models were used; one trained on an unbalanced dataset, InceptionResNet v1 (Sandberg, 2023) and one trained on a balanced dataset, FairFace (Karkkainen & Joo, 2021). The specifications for gender outputs used were shared by the pre-trained models (Male, Female). The FairFace model contained additional parameters for up to 7 races (White, Black, Latino_Hispanic, East Asian, Southeast Asian, Indian, Middle Eastern) and age (0-2, 3-9, 10-19, 20-29, 30-39, 40-49, 50-59, 60-69, 70+) that were not shared in the InceptionResNet v1 model. Both models contained parameters for 4 races (White, Black, Asian, Indian). Age and race variables were not included in measured outputs for the present study. Code using the sex-based language of "Male" and "Female" was changed to the gender-based language of "Man" and "Woman." This shift from sex-based language to gender-based language followed recommended inclusive language guidelines to increase specificity on measured gender identity constructs, as sex at birth was not collected for this study (Bamberge & Farrow 2021). Because binary gender identity labels were used, non-binary transgender participants were excluded from the present study.

Code for both models were taken from the respective GitHub repositories for FairFace and InceptionResNetv1 (IRNv1). Some sections of code were modified from the original repository, including file path, file name, and gender language changes. Unlike the FairFace model, the IRNv1 model was standalone, and did not provide any code infrastructure for classification parameters, testing, and deployment. Thus, original code was written for the IRNv1 model's classification outputs and predictions. The FairFace model was trained on the FairFace dataset, consisting of 108,501 images from the YFCC-100M Flickr dataset. The faces were balanced across 7 defined races. The IRNv1 model was pre-trained on the VGGFace2 dataset, which consists of over 3 million images. The number of men reported within the dataset is approximately 59.7% , with no reports on racial demographics within the original paper (Cao et al., 2017).

Datasets

Because there are no known datasets that are available on the internet consisting of transgender people's faces, the Instaloader Python API (Instaloader, 2023) was used to scrape images of transgender and cisgender people from Instagram. An original Python script was created and used the API in order to download necessary images from target demographics. Through this script, a hashtag was inputted as a string for the scraper, so that all images within the inputted hashtag would be downloaded. Different hashtags were used for cisgender populations and transgender populations. At the time when the images were scraped, all individuals self-identified as either cisgender or transgender.

For transgender populations, hashtags included "#GirlsLikeUs," "#FTMtransgender," etc. The scraped images were then cleaned accordingly (images consisting of subject matter other than binary transgender individuals faces were omitted), and then the testing datasets were thus created. The original images scraped using Instaloader were separated into different folders based on demographics. The images from the datasets were found and used based off of public information available on Instagram, in accordance with Instagram and Instaloaders privacy regulations and guidelines.

For cisgender populations, hashtags were used from racial self-identification. Hashtags used to find cisgender populations on Instagram using the scraper script include "#BlackBoyJoy," "#Latinoman," etc. Categorization of images into folders were done by racial self-identification (7 race categories) rather than gender (as done with *transgender* men/women) to denote that they were in the cisgender dataset. Images scraped using Instaloader were reviewed for accuracy to avoid hashtag overlap between groups (e.g., a transgender Black man who used both "#FTMtransgender" and "#BlackBoyJoy" would have been sorted into the transgender dataset due to gender status).

Images were then renamed post-sorting, removing any personally identifying information (PII) . For example, a transgender man's image name was "TM1_1", indicating that it is the first Trans Man (TM) in the set, and the first image of said man. Numeric identifiers were used to track the number of unique faces in the datasets[7]. If the same person (TM1) had multiple photographs in the dataset, the identifier would reflect the number of instances of their face using an underscore ("TM1_2"). Naming conventions were used in analyses to preserve the original gender self-identification to contrast model-predicted gender identification.

[7] Each different demographic was given a different shorthand signifier. Transgender women would be named as TW#_#. In the cisgender dataset, Latino men would be LM#_#, and Black women would be BW#_#.

Table 1. *Total Number Of Images Per Dataset*

Datasets	N Women	N Men
Transgender (n = 414)	222 (53.62%)	192 (46.38%)
Cisgender (n = 550)	280 (50.91%)	270 (49.09%)

Note. Analyses accounted for incongruent sizes of transgender and cisgender datasets.

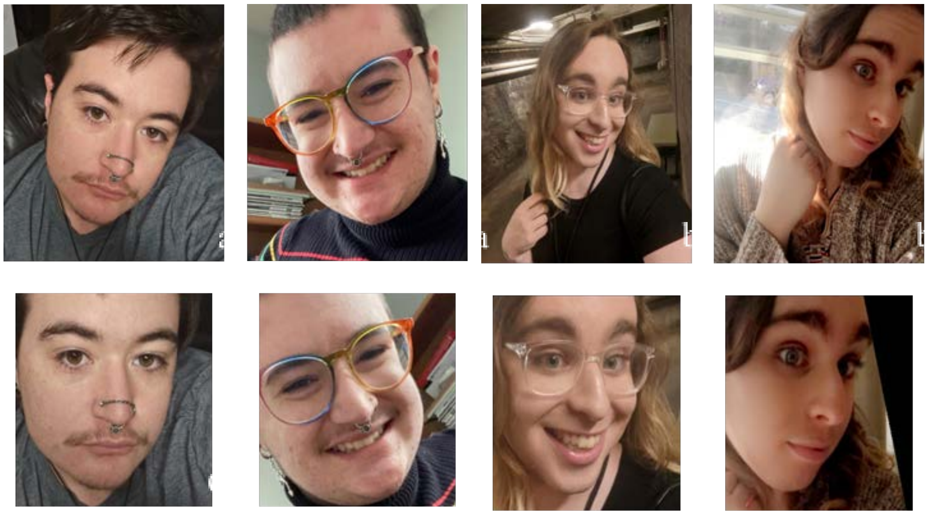
Experiment and Analysis

Both datasets were sorted into comma separated value (csv) files created with image paths. These csv files were used as model inputs, drawing images from the respective file paths of the datasets. A novel python script was created to read through image dataset folders and automatically create csv files for each corresponding dataset. There was one csv file for each dataset (cisgender, transgender), with all image paths to be fed into the neural network. Transgender and cisgender csv files were run separately to maintain congruence with inputs and outputs during testing and deployment.

After inputs were provided to each model, images were cleaned and cropped around faces found within the files and automatically sorted into a folder of "detected" faces, with no supervision or action required on the part of the researcher. After faces were detected, the models made predictions classifying gender identity (man, woman), exported to an output csv file. Outputs from four total csv files (cisgender FairFace, transgender FairFace, cisgender IRNv1, transgender IRNv1) were used in analyses to probe any existing biases within the models.

Within the two specific datasets scraped for this study, there were varying Ns between demographics. Table 1 shown below shows the exact number (N) of images within each dataset created. The exact number of images that were used within each dataset is reported, as well as the rounded percentage that each gender identity (men and women) account for within the whole dataset.

Figure 1. *Examples Of Dataset Inputs To The Fairface And Irnv1 Models*



Note. The two sets of images to the left and to the right were taken from the author and another consenting participant. **a)** and **b)** denote 2 different input images fed into the neural network. **c)** and **d)** denote the same images that were transformed using "*dlib"* to crop around any face detected within the image.

Preliminary analyses of the gender accuracy rates were conducted in a spreadsheet[8] through a formula traditionally used in accuracy calculations for machine learning (Google Developers, 2023):

$$\frac{True\ Positive + True\ Negative}{True\ Positive + True\ Negative + False\ Positive + False\ Negative}$$

Within the context of this study's neural network model classifications, the output was binary and thus only required the calculation of True and False Positive ratings to calculate the accuracy rates. For individuals whose images were included in the test set of this study, their self-identified gender identity was used to determine rating (e.g., a transgender woman classified as a woman True Positive). True and False Negatives are not compatible with the binary classification ratings, thus were omitted from the accuracy calculation. An adapted version of the Confusion Matrix calculator was used in analyses:

[8] Calculations that were conducted in a spreadsheet were also confirmed using Confusion Matrix calculations in RStudio.

$$\frac{True\ Positives + False\ Positives}{N}$$

In the above adapted formula, the True Positives represent the correctly classified individuals, the False Positives represent the misclassified individuals, and N represents the total number of images within each dataset.

Further analysis on the outputs across datasets and models were conducted using R Studio and the R programming language. A logistic regression was performed to determine analysis of each predictor variable on the outcome variable. Predictor variables within analyses were model (FairFace vs InceptionResNetv1 or IRNv1), gender identity (man vs woman), and gender status (transgender vs cisgender self-identification). The sole outcome variable measured are accuracy rates.

Results

Both Models

Unless otherwise specified, gender classification rates only include gender identity. General accuracy rates were calculated for both gender status and gender identity per model and dataset (Table 2). A logistic regression was calculated to determine the odds ratios and effects between both models with respect to the gender identity and gender status on accuracy rates. There was a significant main effect of the model employed, gender status, and gender identity. In regards to the models, it was found that FairFace was 6.27 times more likely to be accurate on general gender classifications than the IRNv1 model, $p < .001$, 95% CI [-.30, -.18]. Based on the gender status, the odds of a cisgender person being gendered correctly was 12.34 times more likely than a transgender person, $p < .001$, 95% CI [-.47, -.33]. Lastly, with gender identity, it was found that the odds of a woman being gendered correctly was 2.73 times more likely than a man, $p < .001$, 95% CI [-.02, .11].

Table 2. *Accuracy Rates per Demographic*

	Datasets		Demographics	
	Gender Status	Total Accuracy	Accuracy Men	Accuracy Women
IRNv1	Cis	68.7%	69.5%	67.5%
	Trans	60.6%	47.4%	72.2%
FF	Cis	95.3%	93.5%	97.5%
	Trans	66.7%	53.6%	77.9%

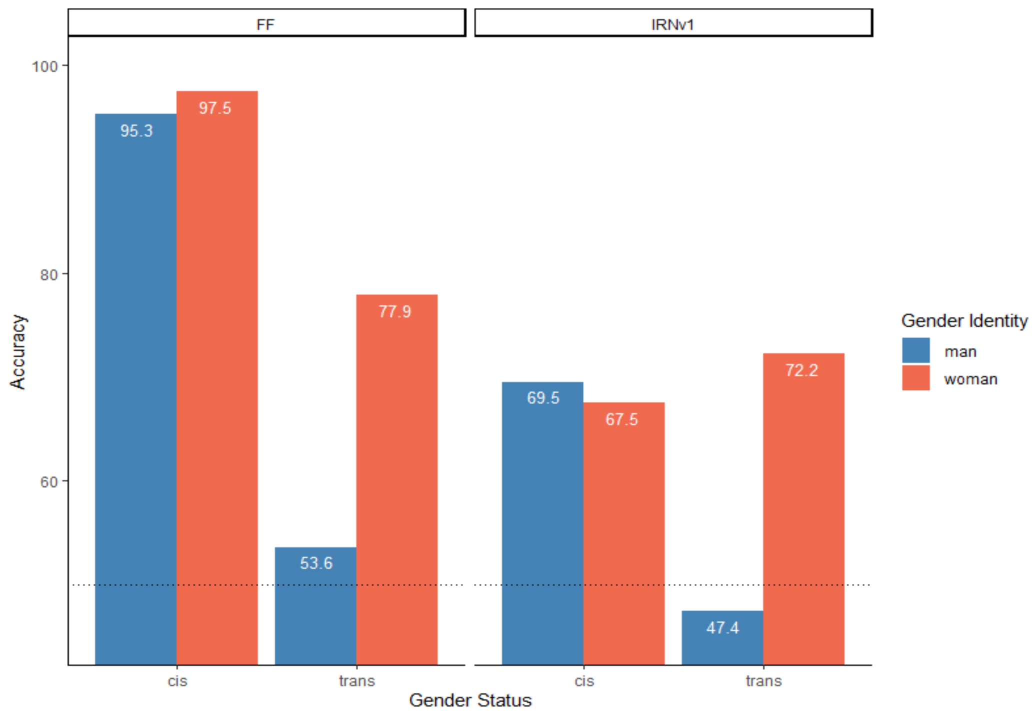
Note. FF and IRNv1 on the vertical axis of the table represent models (FairFace and InceptionResNetv1). Gender status variables were shortened from cisgender and transgender to “cis” and “trans.” The highest accuracy percentage is for cisgender women within the FairFace model (93.5%), and the lowest accuracy is for transgender men within the IRNv1 model (47.4%).

In the same logistic regression, two out of four interactions were found to be significant. The two significant interactions were between model and gender identity, as well as model and gender status. FairFace was found to have a higher accuracy on men, with the odds of a man being gendered correctly as 2.99 times higher than within the IRNv1 model, $p = .05$, CI [-.15, .03]. Additionally, FairFace was more likely to gender a cisgender person correctly around 4.89

times more than the IRNv1 model, p < .001, CI [.07, .28]. The interactions between gender status and gender identity (1.12, p = .82, 95% CI [.10, .30]), as well as model, gender identity, and gender status were not found to be significant (2.83, p = .1, 95% CI [-.08, .21]).

Within a logistic regression, Odds Ratios also serve as the effect size for each outcome. The largest effect on accuracy outputs was from gender status (an Odds Ratio of 12.34 for the gender status alone on accuracy rates). The smallest effect size noted was from a statistically insignificant interaction between gender status and gender identity (an Odds Ratio of 1.12 for the interaction between gender status and gender identity).

Figure 2. *Examples Of Dataset Inputs To The Fairface And Irnv1 Models*



Note. The bar graph is separated between Model - FairFace (FF) and InceptionResNetv1 (IRNv1), Gender Status - cisgender (cis), transgender (trans) testing dataset, and Gender Identity - man (blue / left bars), woman (pink / right bars). The dotted line denotes the chance-level accuracy for classification (50%). Only transgender men had a classified accuracy under 50%.

FairFace Model

For the FairFace model, the binary transgender testing dataset showed an combined accuracy rate of 66.7% for transgender status, regardless of gender identity. Transgender women had an accuracy rate of 77.93%, compared to a 53.65% accuracy rate for transgender men. This is in stark contrast to the cisgender dataset, where classification rates for cisgender women had a 97.5% accuracy rate, and a 93.5% accuracy rate for cisgender men. The combined accuracy rate for cisgender status was 95.3%.

A logistic regression was conducted to determine the FairFace model's correct rate of gender identity classification for 'woman' compared to 'man'. When holding gender status constant, the odds of a woman being gendered correctly within the FairFace model was 2.73 times higher than a man being gendered correctly, $p = .05$, 95% CI [0.18, 19.95]. With gender status, while holding gender identity constant, the odds of a cisgender person being gendered correctly was found to be 1.08 times higher than transgender people within this model, $p < .001$, 95% CI [-3.06, -2.01]. The interaction between gender status and gender identity within the FairFace model was not found to be significant (1.12), $p = .82$, 95% CI [-.92, 1.04].

InceptionResNetv1 Model

Overall accuracy rate on the binary transgender testing dataset was 60.7%. The accuracy rates within the IRNv1 model predictions for transgender women was 72.2%, compared to 47.4% for transgender men.[9] The overall accuracy rate on the cisgender test dataset was 68.6%. Cisgender men had an accuracy of 69.5%, while cisgender women had an accuracy of 67.5%.

A logistic regression was conducted to determine the IRNv1 model's correct rate of gender identity classification for 'woman' compared to 'man'. There was a significant main effect of gender status and interaction between gender status and gender identity. When holding gender identity constant, the odds of a cisgender person being gendered correctly within the IRNv1 model was 1.47 times higher than a trans person, $p < .001$, 95% CI [-1.32, -.58]. The odds of a transgender woman being gendered accurately within this model was 3.22 times higher than a cisgender man, $p < .001$, 95% CI [.64, 1.71]. Transgender gender status created an effect in gender identity that was not present outside of the interaction; there were no significant differences in accuracy rates between gender identities alone (2.47), $p = 0.56$, 95% CI[-.45, .24].

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[9] Transgender men had the lowest average accuracy rate across both models, at 47.7%. Cisgender women held the highest average accuracy rate of 98.2% within the FairFace model.

Discussion

NN Model Outcomes

The results of this study provide important insights into the accuracy of gender identity and gender status classification. The two models within this study were chosen because of the datasets used within pre-training. The creators of the FairFace model have made claims that their dataset used for training was balanced on race, age, and gender; there are no such claims for the IRNv1 model[10].

Overall, both models were typically better at gendering women (regardless of gender status),[11] however with regard to gender status alone, cisgender people were more likely to be gendered correctly than transgender people. The FairFace model was significantly more accurate in general gender classification (both gender status and gender identity) than the InceptionResNetv1 model. Further analysis revealed significant interactions between all predictor variables - model used, gender identity, and gender status. The FairFace model may have deficits in accurately classifying gender for transgender individuals, particularly for trans men. Generally, the FairFace model's accuracy rates for gender classification were influenced more by the gender status (cisgender vs. transgender) rather than gender identity (man vs. woman). The model's accuracy in classifying gender status was not significantly impacted based on gender identity. The IRNv1 model had a larger discrepancy between gender identity and gender status accuracy rates. Cisgender faces were more accurately classified than transgender faces, regardless of gender identity.

Interestingly, the findings from this study echoed similar research previously done by authors on the subject of facial recognition software classifying transgender individuals - with transgender women often being misgendered less than transgender men (Scheuerman et al., 2019). Both between and within models, transgender accuracy rates were less than cisgender accuracy rates. Results were similar in this study, with main effects on gender status from the logistic regressions within and across models. Across both models, cisgender faces were 12.34 times more likely to be gendered correctly than transgender people. Interpretation of the 'effect size' of the Odds Ratio in a similar fashion to Cohen's *d* was done following calculations from a

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[10] The creators of the VGGFace2 dataset, which IRNv1 was pre-trained on, claimed that their dataset was more balanced on gender (men and women) than previous datasets with comparable size.

[11] There is one exception to this generalization across models. In the IRNv1 model, cisgender men had a 2% higher average accuracy rate than cisgender women.

2010 study (Chen et al.). For the IRNv1 and FairFace models, there was a small effect of gender status on accuracy rates. Across both models, there was a large effect of gender status on accuracy rates. The interaction between model and gender status also had a large effect on accuracy rate outcomes.

Biases can exist in neural network models, regardless of the debiasing precautions used (i.e., balancing a dataset). Although FairFace was technically better on all gender classification tasks than the InceptionResNetv1 model, there were still discrepancies pertaining to gender status classification. Despite claims of balancing in the FairFace dataset, gender identity was solely considered in cisgender status,[12] which may have contributed to the observed lapse in accuracy on the transgender dataset. Within both models, transgender accuracy rates were poor compared to cisgender accuracy rates, especially in transgender men[13], who were misgendered the most frequently across both models. We can see that FairFace performed extremely well on cisgender people, with the best average accuracy rates of the datasets and models included. The cause of poor accuracy rates in the "man" gender identity across both models is unknown. Further research is needed to understand the discrepancies between gender identity and gender status, as well as mitigation procedures for these issues.

Study Limitations

There were several limitations within this study, including unbalanced sizes of datasets used to test the NN algorithms, race and age exclusion, and tangible bias mitigation measures. The novel datasets that were created using a Python API scraper contained uneven amounts of images based on gender identity and gender status. The cisgender dataset was more "balanced" with gender categories than the transgender dataset[14], and the number of images between datasets was also uneven.[15] Race and age were also not considered as factors in analysis, and could serve as confounds impacting the accuracy of results on gender identity and status. Within the IRNv1 model, age and race were not able to be programmed correctly at the time of gender analyses,

[12] Within the FairFace paper, there was no mention of the inclusion of transgender people within their attempts to create a balanced dataset. Thus, it is assumed that only cisgender faces were used.

[13] Although transgender men had the worst accuracy rates, both models generally performed worse on men within both gender status groups.

[14] The cisgender dataset had around a 1% difference in men vs women, and images were gathered based on racial groups. The transgender dataset had around 7.24%, and images were not balanced on race.

[15] The number of images within the cisgender dataset was 550, and the number of images within the transgender dataset was 414 - a difference of 136 images or 28.22%.

and thus were omitted from all calculations and predictions used within this study. Additionally, this study did not include specific bias mitigation measures that were recommended by several researchers within the field of ML. Specific bias mitigation measures include modifying gender variables as continuous and using bias detection programs such as InsideBias (Hamidi et al., 2018; Serna et al., 2021). There are major implications of asserting the binary gender distinctions of male/female or man/woman, especially when including transgender individuals. Overall, the most intrusive study limitation was time, causing an inability to pursue the study through the original intended scope.

Future Work

As suggested by Hamidi et al. (2018), the incorporation of gender identity and presentation as a spectrum would be beneficial in the event of inclusion of gender-diverse individuals such as non-binary individuals. Moving away from a dichotomous view of gender would not only allow for the incorporation of non-binary individuals, but would also create more ethical algorithms towards any individual who does not have a gender presentation strictly residing within societal gender norms. Lastly, by classifying gender on a spectrum, we could better identify both human and AI bias with respect to masculinity and femininity, and potentially gender as a whole.

We must include of transgender individuals in datasets to train ML algorithms. Transgender people are already misgendered in real-world contexts due to human bias, and it would be irresponsible to further incorporate algorithmic bias against this already vulnerable population. By including transgender people's faces in facial recognition training datasets, we can potentially mitigate gender misclassification as observed within this study. More studies should be done on the opinions of marginalized groups (such as transgender and gender-diverse individuals) on the ethics and potential uses of such technology, similar to the study conducted by Scheuerman et al. in 2019. When considering the ethical implications of the uses and misuses of any technology, we need to consider the opinions of the people who would be most affected (i.e., marginalized groups such as transgender people, non-white individuals, and a diverse range of ages).

It is also recommended to include more checks for biases within NN algorithms during the training and validation phases in order to potentially check such biases earlier on. Biases, regardless of dataset demographics, should be mitigated through strict procedures and guidelines. Creating tangible rules and guidelines to detect and mitigate biases within NNs, especially those used commercially or through the government, is critical.

Conclusion

Although dataset balancing can provide some benefits in regards to gender classification amongst gender identities, NN classification based on gender status is still lagging behind. The findings from this study highlight the urgent need for further research and development of AI models that are sensitive to the nuances of gender, especially in regards to gender classification. Additionally, we must critically examine the underlying biases and prejudices that may be ingrained in these models, and work to address and mitigate them. This research also underscores the importance of diverse and inclusive datasets for training AI models, as biased data can lead to biased outcomes.

As AI continues to play an increasingly prominent role in our lives, it is crucial that we strive to ensure that these systems are fair, equitable, and just. If biases within NN algorithms are left unchecked, marginalized groups may be severely affected within real-world applications using AI. By recognizing and addressing biases in NN models, we can move towards a more inclusive and equitable future for all individuals, regardless of characteristics such as gender identity, gender status, or gender expression.

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